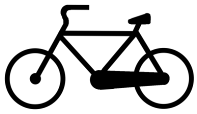


Decision Making

Am I Going to Bike or Walk

Name: ____ Date: ____________

Exercise 1: How do you Decide

1. How did you get to here today?

2. Did you make a decision?

3. What helps you make decision?

4. Do you sometimes make the wrong decision? Give an example

5. Are you sometimes unable to decide?

6. Is it better to decide slowly or quickly?

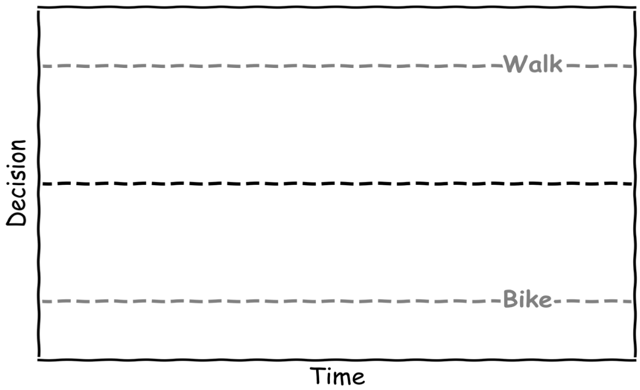
7. What would make you change a decision?

8. What happens when times runs out?

Making a Decision

Exercise 2: A plot is worth 1000 words

Put a start point and explain the different lines on the decision graph:



Exercise 3.1: The decision formula?

The simplified decision formula is the formula of a line:

$$x = \mu t + 0$$

What does the drift rate μ correspond to

a. The slope
b. The intercept
c. The y-variable

Exercise 3.2: Drift to a Decision

If the decision time is 1.5 seconds and the bike threshold is -1 calculate the Drift Rate

$$\mu = \frac{Bike\ Threshold}{Decsion\ Time} = \boxed{}$$

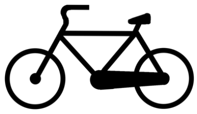
Exercise 3.3: Evidence in Decision

Given the current decision evidence, x(t), is 0.2 and the drift rate is 0.5 calculate the updated decision using the formula

$$x(t+1) = x(t) + 0.5x(t).$$

Answer: $\boxed{}$





Exercise 4.1: Let's make a decision

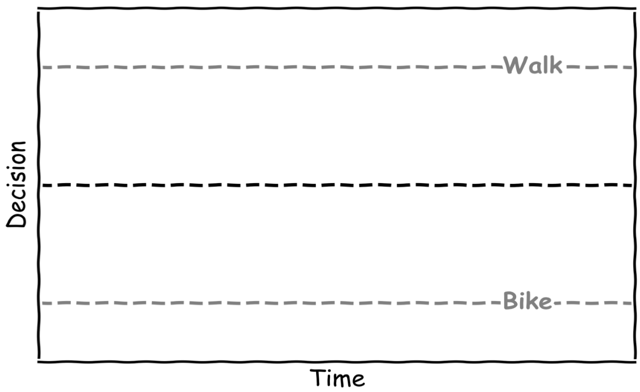
Given the current decision evidence, x(t), is -0.2 and the drift rate is 0.75 calculate the updated decision for two time steps using the formula:

$$x(t+1) = x(t) + 0.75x(t)$$

Time step	Current Decision x(t)	Updated Decision x(t+1)
0	-0.2	-0.2+0.75*(-0.2)
1		
2		

Exercise 4.2: Let's make a decision

Sketch the answers in the table on the graph

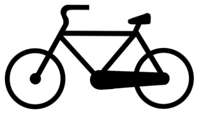


Exercise 4.3: Deciding Fast, Deciding Slow

Will a large drift, μ, mean you decide?

A. Faster

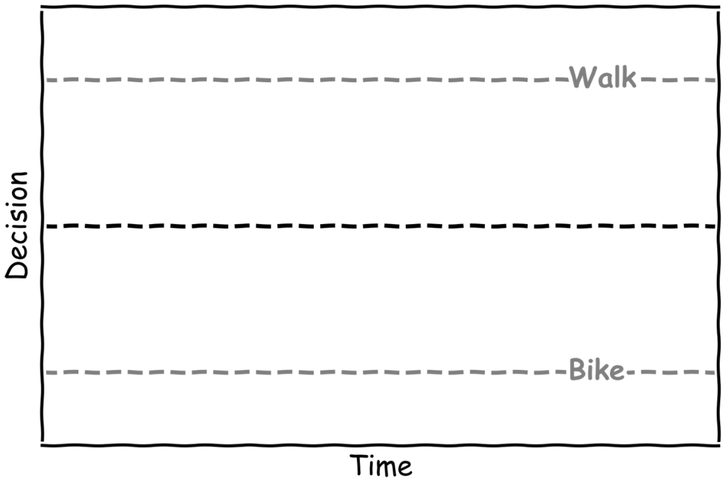
B. Slower

Exercise 5: No Decision

Exercise 5.1: Let's Draw No Decision

Sketch a graph where there is no evidence for the decision:



Exercise 5.2: Let's Calculate no decision

Given the current decision evidence, x(t), is -0.2 and the drift rate is 0.5 calculate the updated decision for two time steps using the formula:

$$x(t+1) = x(t) + 0.5x(t)$$

Time step	Current Decision x(t)	Updated Decision x(t+1)
0	0	0.0+0.5*(0.0)
1		
2		

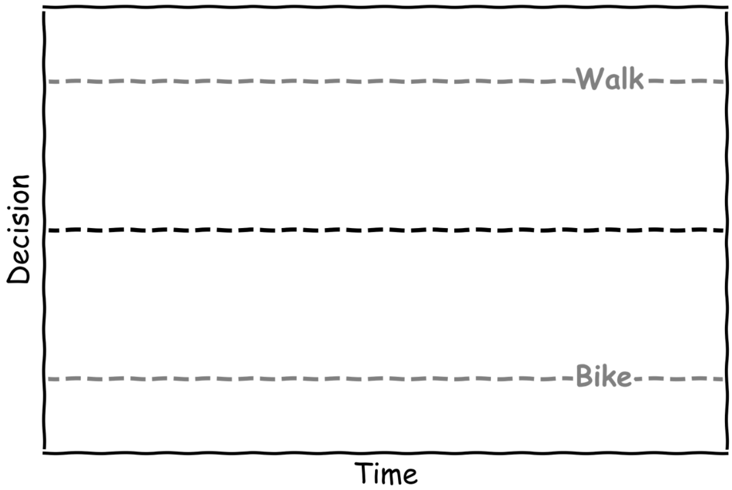
Exercise 5.3: Let's Calculate no decision

What would make it hard to decide to bike or walk?

A. You only love walking
B. You only love biking
C. You love both walking and biking the same

Exercise 6: Let's Bias a Decision

Add a start point to the plot that would be biased to decide to bike:



Given the current decision evidence, x(t), is -0.6 and the drift rate is 0.75 calculate the updated decision for two time steps using the formula:

$$x(t+1) = x(t) + 0.75x(t)$$

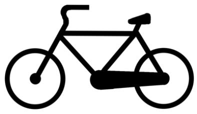
Time step	Current Decision x(t)	Updated Decision x(t+1)
0	-0.6	-0.6+0.75*(-0.6)
1		
2		

Exercise 6.3: Let's Bias Your Decision

What would make you prefer to walk?

A. Better bike paths
B. New bike
C. A friend is walking

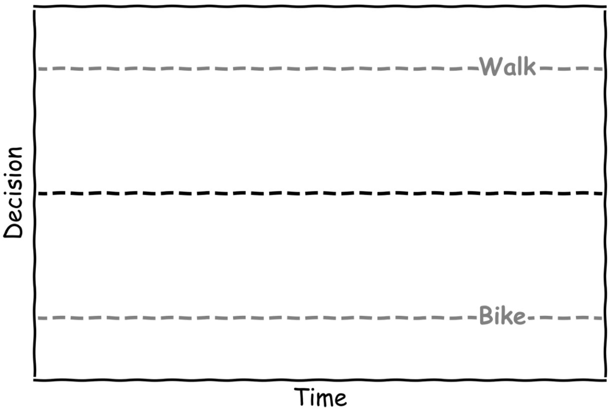




Exercise 7: Let’s Change a Decision

Exercise 7.1: Draw a Change of Decision

Sketch a change of decision going from almost deciding to bike to deciding to walk.



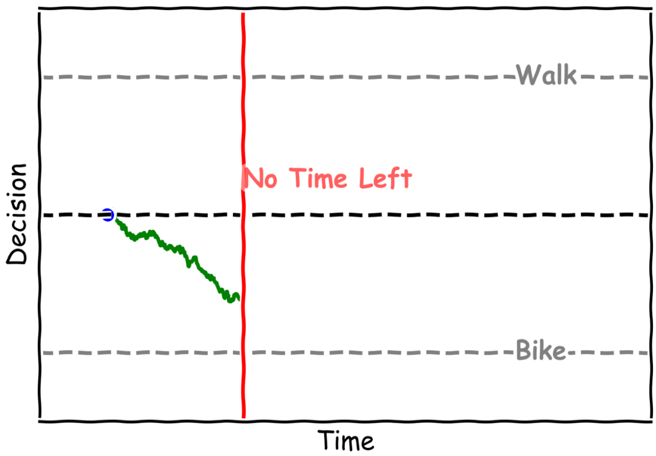
Exercise 7:2: Let’s Change Your Decision

If you were thinking of biking but then you see a “no bike” sign would that change your decision?

A. Yes
B. No

Exercise 8: There Is Not Enough Time

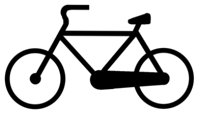
Given the graph below what decision was made when there was no time left?



A. Bike
B. Walk

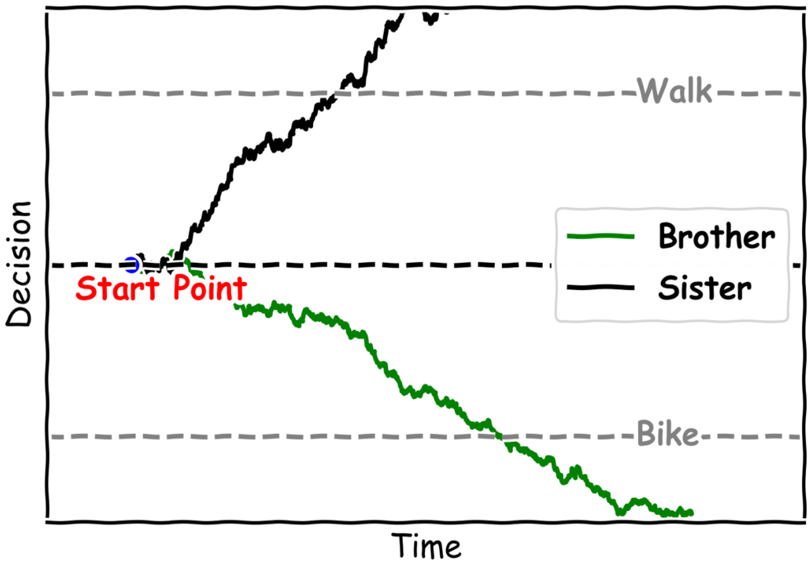
Was it the correct decision?





Exercise 9: Competing Siblings

Given the graph below which sibling got to decide?
How did they travel to school?



Exercise 10: You Decide

Fill in the table below with your own example of a decision making process that could be modelled like biking or walking to school.

Example	What Is the Decision?	What Is the Evidence?	What Affects Speed?
How to get to School	Bike or Walk	The weather or is someone else walking	How much you like walking or biking